

GDG On Campus DMU

REPORT FUMWatch

AI-powered Fire and Smoke Detection



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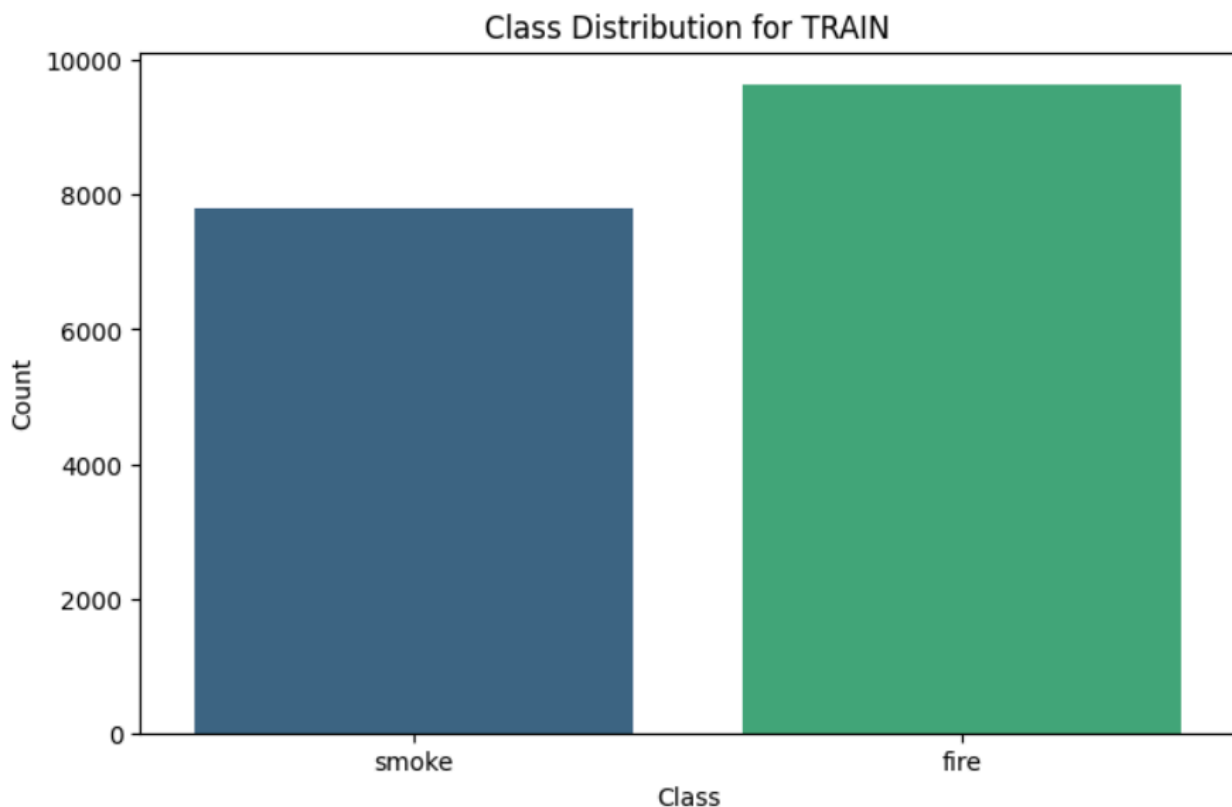
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1.Introduction

Early detection of fire and smoke is critical for protecting lives and property. This study aims to build a comprehensive software system named FumeWatch, using computer vision and deep learning techniques to enable real time detection of smoke and fire sources from images and video streams, while providing an intuitive interactive interface for both cloud and local deployment.

2.Dataset

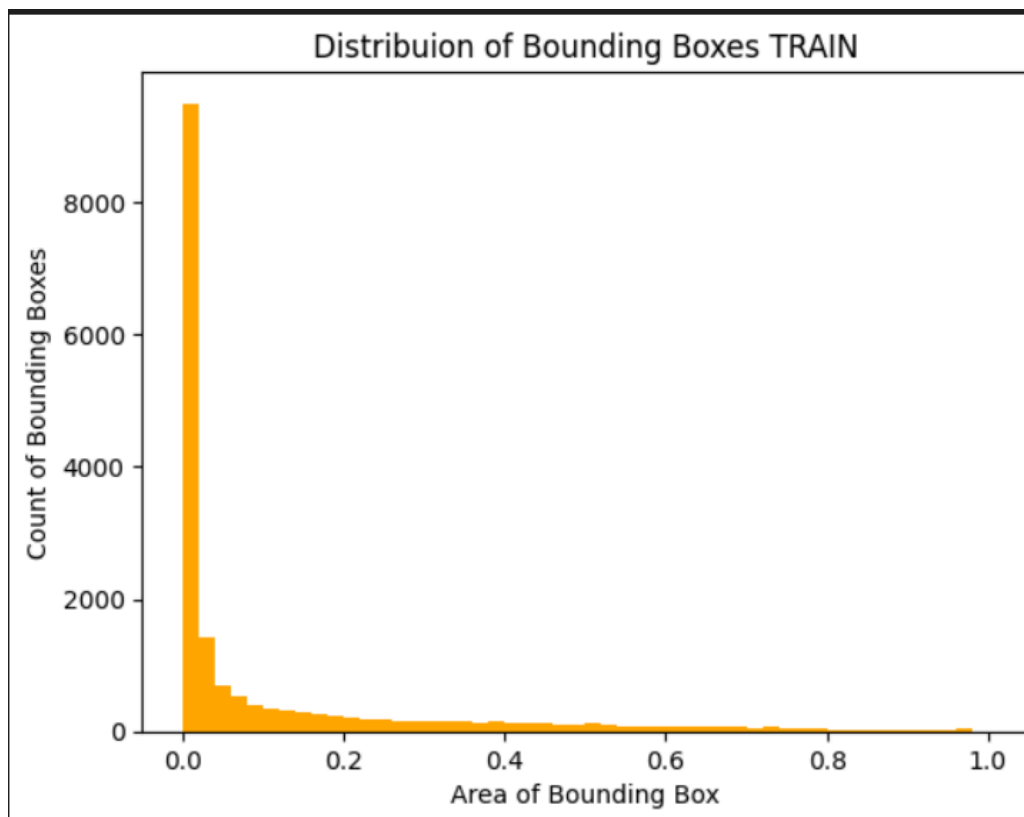
The system is trained and evaluated on specialized datasets designed for smoke and fire detection.and EDA on the data :-



Show distribution of classes between Fire and Smoke for Train.



Sample from Test with its label.



Distribution of bounding boxes around the fire and smoke.

2.1 Dataset Details:-

- Source : Kaggle dataset ([Smoke-Fire-Detection-YOLO](#))
- Classes:
 - Class 0 : Smoke with a blue box.
 - Class 1 : Fire with a red box
- Data Splits:
 - Train: 14,122 IMG
 - Test: 3,099 IMG
 - Validation: 4,306 IMG

2.2 Dataset Preprocessing :-

A custom `preprocess_image` function was implemented in the codebase to enhance model performance:

- **Resizing:** Standardizing all input images and video frames to 640x640 pixels to fit model requirements.
- **HSV Saturation Adjustment:** Converting images to the HSV color space and reducing saturation levels by **40%** to minimize false positives caused.

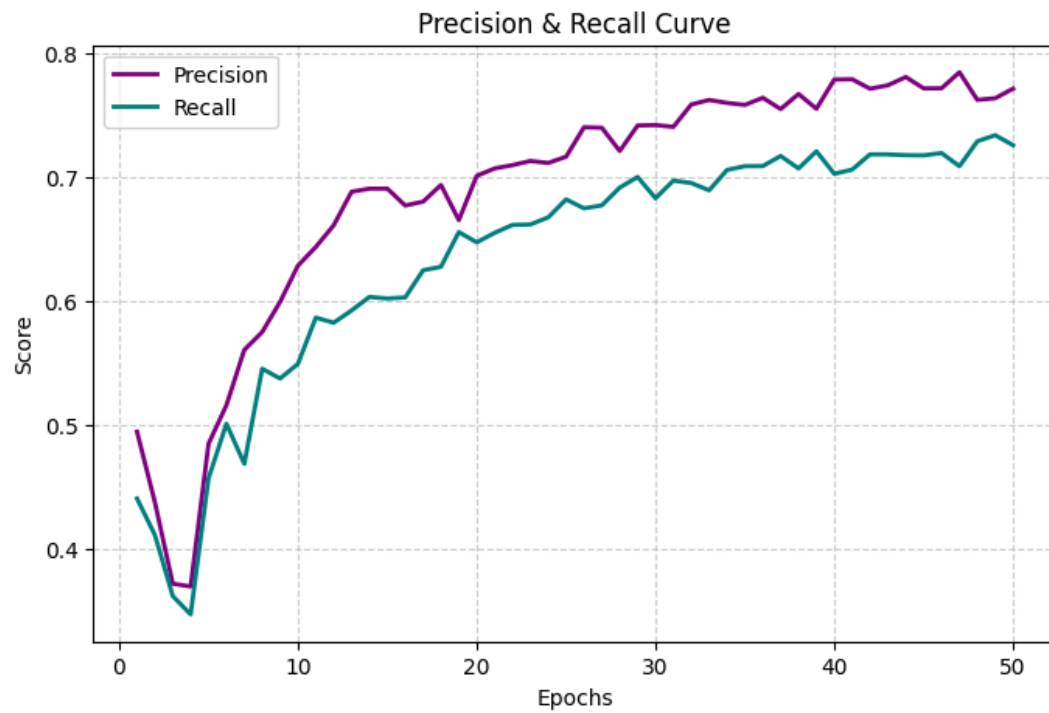
3.Modeling

3.1. Model Details

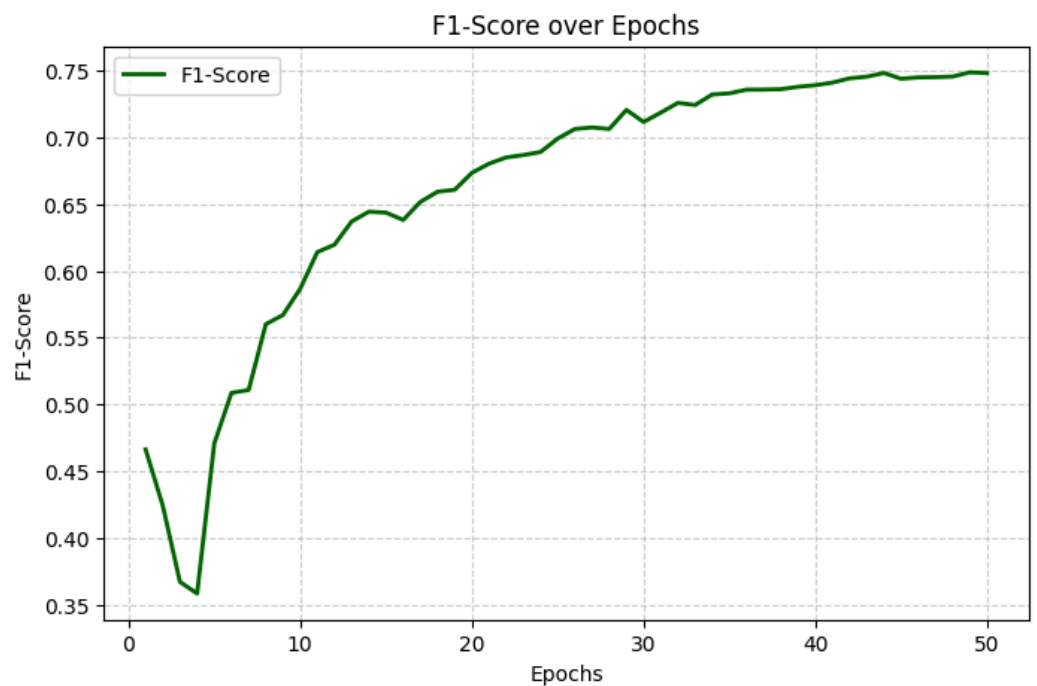
- **Architecture:** YOLOv8.
- **Model Weights:** Custom-trained **weights/best.pt**.
- **Confidence Threshold:** Dynamic threshold, starting at a default value of **0.25**.

Results about Model :

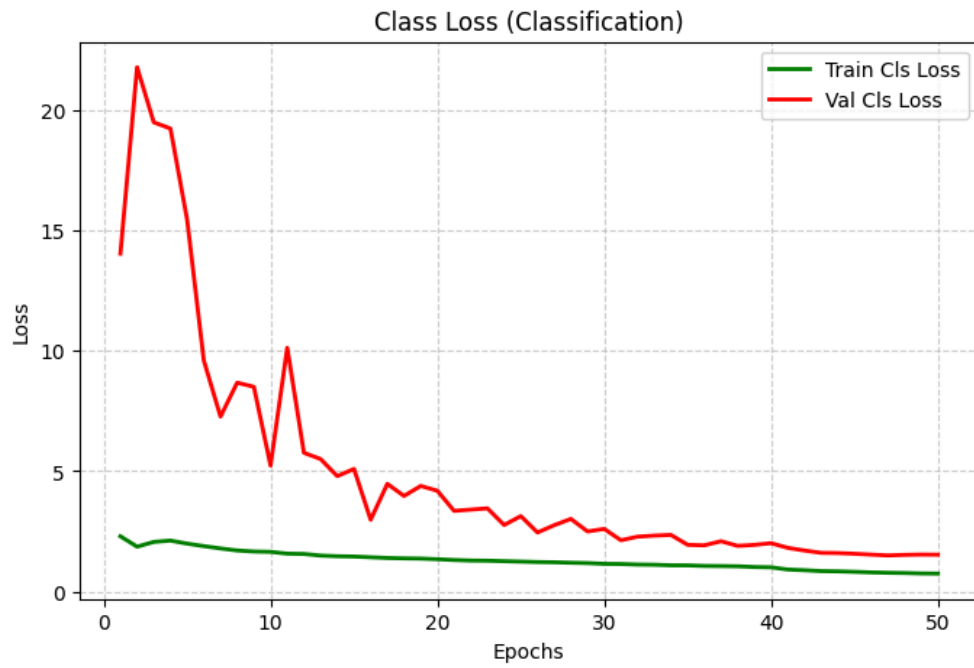
- Precision & Recall



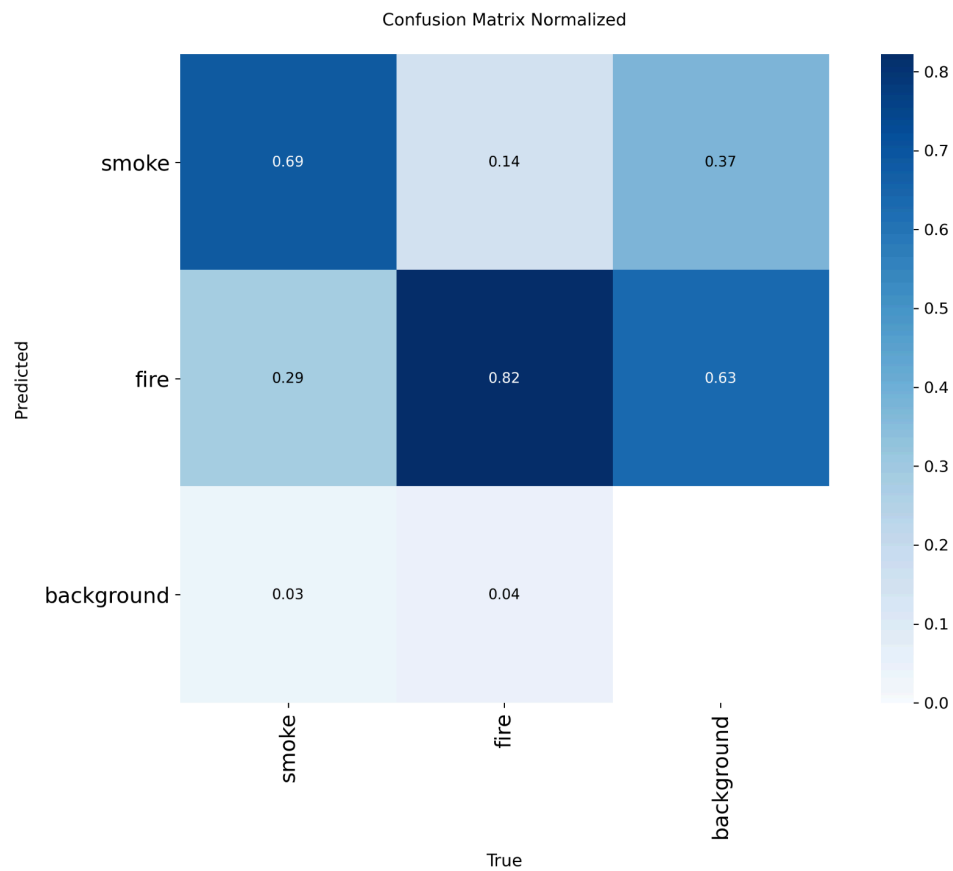
- F1-Score



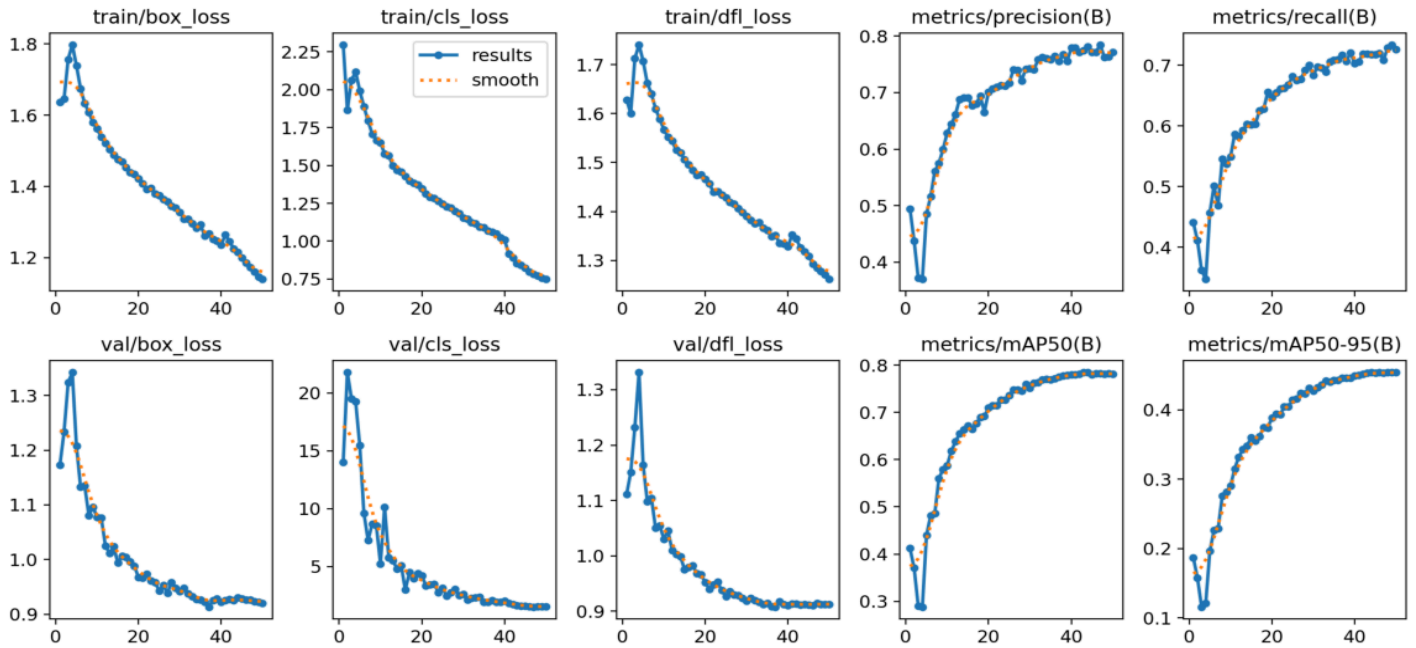
- Class and Box Loss



- Confusion Matrix



- Summary overall Model



4. System Pipeline

The project consists of interconnected software modules designed for efficiency and performance:

1. `app.py`: The user interface responsible for receiving images or videos.
2. `model.py`: Loads model weights then return the model.
3. `utils.py` (`FumeWatchProcessor`): The core execution component responsible for:
 - Preprocessing images (`preprocess_image`).
 - Drawing bounding boxes (`show_box`) .
 - Frame by frame video processing and encoding via `FFmpeg` using the `libx264` codec to ensure playback compatibility across web browsers.

5. Environment & Deployment

The project is structured for easy deployment across various environments:

- **Containerization (Docker):** Built using a `python:3.10-slim` base image, installing required system libraries such as `libgl1` and `libgl1-mesa-glx` to guarantee seamless OpenCV execution.
- **Cloud Deployment:** Pre-configured for direct deployment on Streamlit Community Cloud via custom `packages.txt` files to install `ffmpeg` and `libgl1-mesa-glx` dependencies.

6. Conclusion

The **FumeWatch** project successfully provides an effective and real time solution for detecting fire and smoke. By combining HSV color space preprocessing with YOLOv8 algorithms, the system minimizes false alarms caused by lighting reflections. The Streamlit interface offers a user friendly experience for rapid testing and monitoring in both industrial and residential settings.

U Can Try **FumWatch** From : [FumeWatch - Smoke & Fire Detection · Streamlit](#)

Thanks GDG 